Coffee: A Research-Backed Compendium

From Breeding to Cup — Full Agronomy, Processing, Science & Personalised Experience

Compiled July 2026 · Sources drawn from peer-reviewed journals, WCR, SCA, NCBI/PubMed, and Frontiers

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1. Genetics & Breeding

The Two Commercial Species

Coffee is commercially dominated by two species:

- **Coffea arabica** (Arabica) — ~57% of global production, self-pollinating, relatively low genetic diversity, originating in Ethiopian highland forests. Primary ancestral lines are Typica and Bourbon; disease-resistant introgressed groups include Catimor and Sarchimor.
- **Coffea canephora** (Robusta) — ~43% of global production, cross-pollinating, richer genetic diversity, sourced from selections including Uganda, Congo, Guinea, and their hybrids.

Why Breeding Is Urgent

Developing a pure-line variety from breeding to farmer adoption takes **at least 25 years** (McCook & Montero-Mora, 2024). This lag, combined with accelerating climate change, means breeding decisions made today determine coffee's viability in the 2050s. The challenges include inbreeding depression in Robusta, late expression of target traits, and the need for large trial areas due to the perennial lifecycle.

Current Breeding Programmes

World Coffee Research (WCR) — Innovea Arabica Global Breeding Network WCR operates a collaborative global network now spanning **11 countries** covering 40% of world coffee exports (including Kenya, Uganda, Rwanda, Costa Rica, Mexico, El Salvador). In 2024, WCR launched a first-of-its-kind **speed breeding programme** for arabica, using genomic tools to compress selection timelines. The best trees from the current Innovea population are targeted for commercial testing from **2030**. The International Multilocation Variety Trial (IMLVT) has been extended 6 years at 7 sites to capture yield, disease, and quality performance data post-rejuvenation pruning.

WCR Robusta Breeding Programme (launched 2024) In April 2024, WCR's breeders collected pollen from globally selected parents and initiated the first controlled crosses. Seeds were harvested in early 2025, germinated into seedlings, and are undergoing clonal propagation. The target is **1,000 unique genotypes** (from 44 cross combinations) distributed across network partners — Ghana, Uganda, and Vietnam are confirmed (together producing 48% of global Robusta). Full partner trials begin **2027**; the network formally launches **November 2025**.

Nestlé's Non-GMO Robusta Programme Independently, Nestlé has developed two non-GMO Robusta varieties yielding up to **50% more beans per tree**, alongside documented **30% lower emissions** per unit of production.

Genomics

Whole genome sequences for both *C. canephora* (Denoeud et al., 2014) and *C. arabica* (Scalabrin et al., 2020, 2024) have been published. These enable genomic-assisted selection (GAS) and marker-assisted breeding, though routine integration into national programmes remains limited. Genomic inbreeding analyses and multi-environment trials are increasingly standard in advanced programmes.

Key issue: Genetic conformity in commercial seed sectors is low. A 2024 study presented at the International Coffee Convention (Mannheim) found that seed sold as a named variety frequently does not match that variety genetically — highlighting systemic quality control failures in seed value chains.

2. Seedling Production & Nursery Science

Why Nursery Quality Is Non-Negotiable

WCR's research consistently demonstrates that nursery quality is the **single upstream determinant of long-term farm productivity**. Poor nursery practices can set a farm's productive potential back by years. Key documented findings:

- Only **fully ripe, deep red cherries** should be harvested for seed; immature green cherries have low germination potential; overripe fruits risk pathogen contamination.
- Mother plant selection must prioritise consistent high yield, disease resistance (especially coffee leaf rust, *Hemileia vastatrix*), and cup quality — free from visible pests or disease.

Nursery Systems

Polybag nurseries are the global standard. The East African guide recommends planting-hole amendment with **one 20-litre basin of decomposed organic manure** per hole. ACIAR (Australian Centre for International Agricultural Research) notes that root quality in polybag systems can vary — deformed roots can limit post-planting survival.

Bare-root / secondary nurseries are cheaper but require greater care at transplanting. In either system, the Philippine Coffee Board recommendation of removing the polybag *entirely* at transplanting is standard practice — never leave plastic around the root ball.

Pest, Disease & Nutrition Management

Integrated Pest Management (IPM) — combining biological controls with selective pesticide use — is WCR's recommended standard. Common nursery pests: aphids, thrips, caterpillars. Common diseases: damping-off, coffee leaf rust. **Phosphorus** is critical at transplanting stage for root establishment.

Vegetative Propagation

For Robusta (which cannot self-fertilise), **grafting** is increasingly used. WCR's "Grafting Robusta" manual (2023) provides step-by-step protocols for both farmer workshops and individual practice. Grafting onto disease-resistant rootstocks can dramatically improve plant health in soils with root pathogens.

3. Growing Conditions — Terroir, Soil & Climate

Altitude

Altitude is the most consistently studied environmental variable in coffee quality research. The evidence is strong:

- Higher altitudes (typically 1,200–1,800 m a.s.l. for Arabica) produce **cooler temperatures, slower cherry maturation**, and higher accumulations of sugars, chlorogenic acids (CGAs), trigonelline, and caffeine in green beans.
- A study in Ethiopia (Springer, *Agronomy for Sustainable Development*, 2022) found that **72.3% of the variation** in total preliminary cup quality could be explained by soil temperature and soil chemical variables — with cool, elevated conditions consistently outperforming warm, low-altitude sites.
- East-facing slopes show superior cup quality attributes in Costa Rican terroir studies, attributed to better morning sunlight exposure and acidity development.

Soil

Volcanic soils dominate premium coffee regions (Central America, Kenya's Kirinyaga County) partly for their drainage and mineral content. Research notes the **lack of strong consensus** on the direct flavour contribution of soil minerals vs. other variables — but soil's role in water retention, drainage, pH, and microbiome structure is well established.

Preferred soil conditions:

- Deep, well-draining, slightly acidic (pH 5.5–6.5)
- Rich in organic matter
- High cation exchange capacity (CEC) for nutrient retention

Microbiome

A 2020 Brazilian study (PMC7477199) demonstrated that altitude significantly structures the bacterial and fungal communities in both coffee soil *and* fruit — and that these indigenous microorganisms influence fermentation outcomes post-harvest. Higher-altitude farms showed greater bacteria–bacteria co-occurrence networks, while lower altitudes showed more fungi–fungi interactions.

Shade & Microclimate

Shade trees buffer temperature, reduce UV stress, and increase humidity. However, research shows **trade-offs**: excessive canopy cover reduces yield and can negatively affect quality by limiting light. Shade management is identified by a major Frontiers review (*Front. Plant Sci.*,

2021) as among the most promising climate-change mitigation strategies for quality preservation.

Climate Change Vulnerability

The same Frontiers review (2021) confirmed that different coffee regions exhibit **geographic quality signatures** (terroir), but found no single region universally more vulnerable than others. The priority management responses are: shade management, climate-resilient cultivar selection, wild coffee germplasm, soil nutrient management, and IPM.

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4. Harvest — Timing & Method

Selective Hand-Picking (Strip vs. Selective)

Research is unambiguous: **selective hand-picking of only fully ripe cherries** produces the highest cup quality. Strip harvesting (removing all cherries from a branch simultaneously) introduces underripe and overripe fruit, degrading fermentation outcomes and elevating defect rates.

Maturity Indicators

Fruit colour (red or yellow depending on variety) is the primary field indicator. Brix measurement of fruit sugar content is used in advanced operations. The research from Colombia (Coffee Science, 2024) showed that post-harvest processing quality is the **dominant factor affecting cup quality (57.6%)**, with farm environment accounting for 37.9%.

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5. Post-Harvest Processing

Processing is the most research-active area of coffee science. It involves removing the fruit layers surrounding the green bean and preparing the bean for roasting. Processing method profoundly determines flavour precursor composition.

Core Methods

Method	Key Characteristics	Flavour Profile
Washed (Wet)	Full pulp removal, fermentation, washing	Bright acidity, clarity, chlorogenic acid/malic acid/sucrose dominant

Method	Key Characteristics	Flavour Profile
Natural (Dry)	Whole cherry dried in sun	Full body, fruity sweetness, fructose and GABA dominant
Honey / Pulped Natural	Skin removed, mucilage retained during drying	Intermediate between washed and natural; highest antioxidant activity
Anaerobic	Fermentation in sealed, oxygen-free environment	Intense, complex, wine-like; high lactic acid accumulation

Research Findings by Method

Washed: Uses pulping machines to remove skin, then natural fermentation removes mucilage (typically 24–72 hours), followed by washing and drying. Parameters including fermentation duration, soaking application, and depulping vs. demucilaging style all significantly alter microbial communities and metabolite composition (NCBI/PMC, 2019 Yunnan study).

Honey/Pulped Natural: Research (ScienceDirect, 2022) confirms lactic acid bacteria (LAB) dominate (48%) during honey processing, with *Wallemia* as the major fungal genus (31%). Extended fermentation to 240 hours produced **15% sweeter** and more consistent cup profiles. Honey-processed coffee shows improved colour, sugars (+8.12%), proteins (+4.89%), and polyphenols (+4.66%) compared to wet-processed. Critically, honey processing uses ~10% of the water of wet processing — a significant sustainability advantage.

Drying Method Impact: A 2024 Brazilian study (*Foods* 14) comparing sun-drying vs. mechanical drying (CoffeeDryer®) found mechanical drying preserved significantly higher phenolic compounds and antioxidant activity (3.24 g EAG/100g vs. 2.20 g EAG/100g in sun-dried Alta Mogiana coffees).

Fermentation as Quality Driver: A 2024 systematic review (Semantic Scholar, 118 articles from ScienceDirect, Scopus, SpringerNature, 2014–2024) confirmed fermentation method is the strongest lever on volatile compound profile and bioactive compound content. Caffeine content in arabica **increased with roasting temperature** following wet fermentation, due to decomposition of acids elevating the proportion of non-liquid substances.

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6. Roasting Science

Roasting transforms green beans through a sequence of chemical reactions: Maillard reaction, Strecker degradation, caramelisation, hydrolysis, and pyrolysis. Temperature, duration, and rate of development (RoR) all determine the final chemical and sensory profile.

Roast Level and Key Compounds

Roast Level	Temperature Range	Characteristics
Very Light / Cinnamon	~198°C	Green, grassy, high acidity, max CGAs, fruity/floral volatiles
Light (American)	~212°C	Floral, fruity, bright acid, high polyphenols
Medium	~220°C	Balanced sweet and roasted; highest ABTS+ antioxidant capacity
Dark (Italian)	~228–230°C	Smoky, chocolate-like; low CGAs; high bitter compounds (phenylindanes, furans, quinic acid derivatives)

Key research findings (2022–2024):

- **Chlorogenic Acids (CGAs)**: Peak in unroasted/light-roasted beans; roasting significantly degrades them. A 2025 systematic review (*Front. Nutr.*) found that 5-O-caffeoylquinic acid (5-CQA), the most abundant CGA, activates the Nrf2 antioxidant pathway. A typical coffee serving contains **27–121 mg of CGAs**.
- **Caffeine**: Chemically stable across roast levels. Despite popular belief, mass per serving is similar across roast levels when dosed by weight.
- **Diterpenes (Cafestol & Kahweol)**: Found in coffee oils, their retention is **highly brewing-method-dependent** (see Equipment section). They have paradoxical effects — anti-inflammatory and potentially pro-atherogenic depending on dose.
- **Acrylamide**: A mildly concerning compound that forms during roasting; peaks at medium roast and decreases at dark roast. Different filter methods significantly alter its content (Antioxidants, 2023).
- **Volatile compound development**: A 2024 study (*Molecules* 29:4723) using SCA guidelines identified 74 compounds at four roast levels, of which 25 were aroma-active.

NIR spectroscopy + chemometrics is now validated as a rapid industrial quality-monitoring tool (Molecules 2024, 29:318).

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7. By-Products & Circular Economy

Coffee processing generates large volumes of by-products at every stage. These are increasingly recognised as high-value functional ingredients rather than waste.

By-Product Map

By-Product	Source Stage	Composition Highlights
Coffee Pulp / Cascara	Wet processing (skin + pulp)	High polyphenols, flavonoids, caffeine (9.20 mg/g), chlorogenic acid, protocatechuic acid
Mucilage	Wet processing	Pectin, sugars, organic acids
Coffee Husk	Dry processing	Pulp + pectin + parchment; ~0.18 t husk per tonne of fruit
Parchment	Hulling	Cellulose, hemicellulose, minerals
Silverskin / Chaff	Roasting	Dietary fibre, polyphenols, melanoidins; caffeine ~5.30 mg/g
Spent Coffee Grounds (SCG)	Brewing	Lipids, proteins, phenolics; largest by volume

Research-Backed Applications

Cascara (dried coffee cherry husk):

- Brewed as a tea/infusion with dominant citric acid, tamarind, and black-tea flavour notes; high antioxidant content.
- 2026 study (Vaher & Bragina, *Molecules* 31:403, Tallinn University of Technology) characterised cascara's **antimicrobial activity against ESKAPE pathogens** — clinically resistant bacterial species — driven by its polyphenol and flavonoid content.

- Shown to function as a **prebiotic** ingredient (Machado et al., *Crit. Rev. Food Sci. Nutr.*, 2024).
- Emerging use as **edible active coating** for food preservation (e.g., hazelnut preservation, 2024).

Silverskin:

- Rich in dietary fibre; used in biscuits, cookies, and baked goods to enhance fibre content and antioxidant properties (Applied Food Research, 2024).
- Ultrasound extraction methods (green extraction technology) show superior bioactive recovery compared to conventional solvent extraction.

Spent Coffee Grounds (SCG):

- Can be incorporated into cakes, cookies, muffins, breads, yogurts, and fermented/distilled beverages.
- Rich in lipids and used as biofuel feedstock, compost, and soil amendment.
- Active research into SCG as a biosorbent for heavy metal removal from water.

Green Extraction Technologies (2026 Annual Reviews, Wiley): Supercritical CO_2 extraction, deep eutectic solvents, and enzyme-assisted approaches are the current frontier — demonstrating superior efficiency in recovering target bioactive compounds while minimising environmental footprint.

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8. Equipment — What the Research Says

The Grinder: The Most Important Variable

Research and professional consensus are aligned: **grind consistency is the single most influential factor** in cup quality for filter/pour-over methods. The grind determines surface area, extraction rate, and fines distribution.

Burr type:

- **Flat burrs** (64mm+): Maximum uniformity; preferred for filter. Lower particle size distribution (PSD) spread.
- **Conical burrs**: Better retention characteristics; suitable for both espresso and filter.

Key research implication: Grind size interacts with roast level and bitterness. A 2024 study (Habara & Horiguchi) found bitterness increases with darker roast AND finer grind size — the two variables compound.

Espresso Extraction

Standard parameters (research-validated): **90–95°C**, **~9 bar** pressure, 25–30 second extraction. These create the characteristic crema and concentrated flavour through forced emulsification of coffee oils.

Ultrasonic Brewing (2026 cutting edge): A 2026 study (*ScienceDirect*, Australian research group) demonstrated ultrasonic cavitation at **42.6 kHz** can achieve espresso-strength extraction in **2–3 minutes at low temperature** with lower energy consumption than conventional espresso. The system was retrofitted into a commercial Ascaso Uno machine. This represents the research frontier for sustainable, low-temperature extraction.

Filter / Pour-Over

Pour-over methods (V60, Chemex, Kalita Wave) produce clarity and accentuate single-origin character. The brew control chart (*J. Food Sci.*, 2023) relating extraction yield (ideally 18–22%), total dissolved solids (TDS: 1.15–1.55%), and brew ratio to sensory properties is the key diagnostic tool for precision brewing.

Automated pour-over is a major 2024–2025 equipment trend at the specialty commercial level, aiming to eliminate barista variability while maintaining the filter quality profile.

Cafestol & Kahweol by Brewing Method

Diterpene retention varies dramatically by preparation method — a clinically relevant distinction:

Method	Diterpene Level
Filtered (paper)	Near zero (paper captures lipids)
French press / plunger	High
Espresso	Moderate
Moka pot	Moderate-High
Cold brew	Variable (medium)

This is clinically significant: unfiltered coffee is associated with small increases in LDL cholesterol. Paper-filtered coffee eliminates this effect while retaining other health benefits.

Temperature Precision

Research supports the extraction sweet spot of **91–94°C** for most Arabica. PID-controlled boilers (Proportional–Integral–Derivative temperature controllers) are the gold standard for repeatability. Thermally unstable machines introduce shot-to-shot variation that masks the effect of all other variables.

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9. Preparing Coffee By-Products

Cascara (Coffee Cherry Tea)

Preparation: Steep dried cascara husks in hot water (70–80°C — lower than coffee, to preserve delicate phenolics) for 4–6 minutes. Use a ratio of approximately 15–20g per 250ml.

Flavour expectation: Cherry, tamarind, hibiscus, mango notes; mild caffeine (~1/4 of a coffee); bright tartness from citric acid; body similar to black tea. Research confirms the sensory profile is distinct from coffee — it sits closer to hibiscus or rosehip infusions.

Fermented cascara (traditional Yemeni *Qishr* style): Cascara + ginger + hot water produces a warming, spiced infusion consumed in coffee-producing regions for centuries.

Cold-brew cascara: Steep at room temperature or refrigerated for 12–18 hours for a refreshing, lower-extraction cold drink. Research shows cold extraction preserves more of the delicate fruit esters.

Silverskin

As a tea/infusion: Steep 5g roasted silverskin in 200ml water at 85°C for 3–4 minutes. The flavour is mild, slightly smoky, and light-bodied. Less common as a beverage; more widely used as a food additive.

As a food ingredient: Incorporate silverskin powder (ground chaff) into baked goods at 5–10% substitution of flour. Research (*Applied Food Research*, 2024) shows this meaningfully increases dietary fibre and antioxidant activity without significantly degrading sensory scores.

Spent Coffee Grounds (SCG)

Culinary uses: Use as a dry rub for meat (coffee crust); incorporate into brownie or cookie batters (1–2 tbsp per batch); add to smoothies with banana and cacao for a bitter, complex base.

Garden use: SCG is a mild nitrogen source and soil conditioner. It does not significantly acidify soil contrary to popular belief (pH is nearly neutral once brewed). Excellent for compost activation.

Green Coffee (Unroasted)

Green coffee extract is studied for weight management (chlorogenic acid content). **Not recommended for direct brewing** — the flavour is harsh and grassy, and extraction of beneficial compounds requires either light processing or supplemental format.

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10. Health & the Individualised Experience

Summary of Research Consensus (2024–2026)

A comprehensive 2026 American Heart Association review concluded that **moderate caffeine or coffee consumption (up to 400 mg/day caffeine; ~3–5 cups of 8 oz coffee) is safe for most adults** and is associated with lower risk of:

- Cardiovascular disease (coronary heart disease, stroke, heart failure, atrial fibrillation)
- Type 2 diabetes
- Liver disease (cirrhosis, fatty liver, liver cancer)
- Cognitive decline / Alzheimer's disease
- Parkinson's disease
- Uterine and skin cancers (melanoma)

A 2024 PubMed review (*Geroscience* 46:6473–6510, Ungvari et al.) confirmed that cardiometabolic benefits of coffee are **consistent across age, sex, and geographical region** and involve antioxidative, anti-inflammatory, lipid-modulating, insulin-sensitising, and thermogenic mechanisms.

Key Bioactive Mechanisms

Compound	**Primary Mechanism**	**Notes**
Chlorogenic Acids	Nrf2 pathway activation; α-glucosidase inhibition	Depleted by dark roasting
Caffeine	Adenosine receptor antagonism	Main psychoactive; half-life 3–7 hrs; genetic variation

Compound	Primary Mechanism	Notes
Trigonelline	Neuroprotection; insulin secretion	Heat-labile; light roast retains more
Cafestol/Kahweol	Anti-inflammatory; paradoxical LDL effect	Removed by paper filtration
Melanoidins	Prebiotic; antioxidant; anti-inflammatory	Form during roasting; increase with roast level

Individualising the Coffee Experience

Caffeine metabolism — CYP1A2 genetics: The enzyme CYP1A2 determines caffeine metabolism speed. "Fast metabolisers" can consume more caffeine without cardiovascular risk; "slow metabolisers" may see LDL and blood pressure effects at the same dose. Genetic testing is now commercially available and informs individualised guidance.

Roast level and health: Light roast retains more CGAs and antioxidant capacity; dark roast has more melanoidins and less acidity. A 2025 study (*Antioxidants* 14:285, PMC11939571) reinforces coffee's anti-aging role through oxidative stress modulation — most prominent in lighter roasts.

Brewing method and individual lipid profile: Those with elevated LDL cholesterol benefit significantly from switching to **paper-filtered** coffee (V60, drip, Chemex) to eliminate cafestol/kahweol. This is a rare case where equipment choice has documentable clinical relevance.

Timing and circadian rhythm: Research (Rashidy-Pour, 2024) supports the widely-practised "cortisol window" approach — waiting 60–90 minutes after waking before first coffee, allowing natural cortisol to peak and decline, improving caffeine's net alertness effect.

Sleep sensitivity: Caffeine's half-life means a 200mg coffee at 2pm delivers ~100mg at 9pm for average metabolisers. For sleep-sensitive individuals, research supports cutting off caffeine before 1–2pm and opting for **decaffeinated** or **low-caffeine** preparations (cascara, low-dose filter) in the afternoon.

Pregnancy: The current evidence-based guidance remains **below 200 mg/day** during pregnancy, and many clinicians recommend decaf for the first trimester.

Gut and microbiome: Coffee's phenolics undergo colonic fermentation (*Food Chemistry*, 2025, Cañas et al.) generating metabolites that interact with gut bacteria. Individual microbiome composition influences how much benefit a person extracts from coffee polyphenols — an active research frontier.

The Sensory Self-Assessment Framework

To build an individualised coffee experience, the research supports this evaluation sequence:

1. **Start with origin** (terroir shapes the foundation; Ethiopian naturals differ fundamentally from washed Kenyan or Colombian)
2. **Select roast level** (light for acidity/complexity; medium for balance; dark for body/bitterness — matched to your palate and health goals)
3. **Choose processing method** (washed for clarity; natural for sweetness; honey for both)
4. **Match brewing method to your lipid profile and taste goals** (filter = clean, low-fat; plunger = rich, high-fat; espresso = concentrated, moderate-fat)
5. **Dial in grind** (the most impactful variable for flavour consistency at home)
6. **Consider timing** (morning cortisol window; cut-off for sleep)
7. **Explore by-products** (cascara in the afternoon as a lower-caffeine, high-antioxidant alternative)

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11. Deep-Research Links by Section

Breeding & Genetics

- WCR Innovea Network: https://worldcoffeeresearch.org/programs/global-breeding-network
- WCR Robusta Breeding (2025 update): https://worldcoffeeresearch.org/news/2025/robusta-breeding-a-year-later
- Frontiers Collaborative Breeding Review (2024): https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2024.1431849/full
- Genomic-Assisted Breeding, *The Plant Genome* (2024): https://acsess.onlinelibrary.wiley.com/doi/10.1002/tpg2.20321
- Climate-Smart Breeding, ScienceDirect: https://sciencedirect.com/science/article/abs/pii/S0065229624000946
- Democratising Coffee Genetics, ICC 2024 (MDPI Proceedings): https://www.mdpi.com/2504-3900/109/1/32

Nursery Management

- WCR Good Practice Guides: https://worldcoffeeresearch.org/programs/guides

- WCR Nursery Management Guide: https://worldcoffeeresearch.org/resources/good-practice-guide-coffee-nursery-management
- Kenya Coffee School, Nursery Module: https://kenyacoffeeschool.golearn.co.ke/kcs-coffee-agronomy-day-3-coffee-nursery-establishment-and-seedling-production/

Terroir, Soil & Climate

- Altitude & cup quality, Ethiopia (Springer, 2022): https://link.springer.com/article/10.1007/s13593-022-00801-8
- Terroir & Agro-Ecosystem (ResearchGate): https://www.researchgate.net/publication/312185631_Cultivating_Coffee_Quality-Terroir_and_Agro-Ecosystem
- Climate change & coffee quality (Frontiers Plant Sci., 2021): https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2021.708013/full
- Microbiome & altitude, Brazil (PMC, 2020): https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7477199/

Post-Harvest Processing

- Honey processing volatiles & microbial diversity (PMC, 2025): https://pmc.ncbi.nlm.nih.gov/articles/PMC11838140/
- Colombia dry fermentation quality comparison (Coffee Science, 2024): https://coffeescience.ufla.br/index.php/Coffeescience/article/view/2163
- Honey/pulped natural metagenomics (ScienceDirect, 2022): https://www.sciencedirect.com/science/article/abs/pii/S1466856422001436
- Wet processing parameters & microbiology (NCBI, 2019): https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6863779/
- Drying methods & quality (NCBI, 2025 — 2024 harvest data): https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12071231/
- SCA Fermentation Effect overview: https://sca.coffee/sca-news/25-magazine/issue-10/english/the-fermentation-effect-25-magazine-issue-10

Roasting

- Roast level & aroma-active compounds (Molecules 29:4723, 2024): https://www.mdpi.com/1420-3049/29/19/4723
- NIR spectroscopy for roast monitoring (Molecules 29:318, 2024): https://doi.org/10.3390/molecules29020318
- Roasting & phenolic/volatile compounds (Wiley FSN, 2022): https://onlinelibrary.wiley.com/doi/10.1002/fsn3.2849

- Roast level bioactive & aromatic compounds (Int. Agrophys., 2024): https://www.international-agrophysics.org/Effect-of-the-roasting-level-on-the-content-of-bioactive-and-aromatic-compounds-nin,176300,0,2.html

By-Products

- Cascara antimicrobial against ESKAPE (PMC, 2026): https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12898695/
- Coffee by-products as functional ingredients (Annual Reviews, 2026): https://www.annualreviews.org/content/journals/10.1146/annurev-food-052924-114847
- Upcycling by-products review (Wiley CRFSFS, 2026): https://ift.onlinelibrary.wiley.com/doi/10.1111/1541-4337.70401
- Circular economy upcycling (MDPI Processes 12:2851, 2024): https://www.mdpi.com/2227-9717/12/12/2851
- Bioactive compounds, cascara & silverskin: https://www.researchgate.net/publication/355950070_The_assesement_of_bioactive_potential_and_sensory_acceptability_of_coffee_and_its_byproducts-_cascara_and_silverskin
- Coffee by-products in bio/edible films (Frontiers, 2025): https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2025.1667255/full
- Bioactive compounds, Mexico origin study (PMC, 2024): https://pmc.ncbi.nlm.nih.gov/articles/PMC11478550/

Health & Individualised Experience

- AHA-level cardiovascular review (Geroscience 46, PubMed 2024): https://pubmed.ncbi.nlm.nih.gov/38963648/
- Frontiers — Coffee as targeted nutritional intervention (2025): https://www.frontiersin.org/journals/nutrition/articles/10.3389/fnut.2025.1690881/full
- Coffee as antioxidant elixir (Antioxidants 14:285, PMC, 2025): https://pmc.ncbi.nlm.nih.gov/articles/PMC11939571/
- NCA health benefits overview (with 2026 AHA statement): https://www.aboutcoffee.org/health/health-benefits-of-coffee/
- National Center for Health Research — benefits vs. risks: https://www.center4research.org/coffee-health-benefits-outweigh-risks/
- MDPI Nutrients — coffee impact on health & wellbeing (2025): https://www.mdpi.com/2072-6643/17/15/2558
- News-Medical — molecular mechanisms (2025 Frontiers Nutrition): https://www.news-medical.net/news/20251119/Coffee-protects-the-brain-metabolism-and-immunity-at-the-molecular-level.aspx

Equipment Science

- Ultrasonic espresso extraction (ScienceDirect, 2026): https://www.sciencedirect.com/science/article/pii/S0260877426002311
- Brew control chart (J. Food Sci. 88, 2023): doi.org/10.1111/1750-3841.16531
- Innovative filter cups & bioactive benefits (ScienceDirect, 2024): https://www.sciencedirect.com/science/article/pii/S2666154324000619

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